

Work Experience

Data Scientist & AI Engineer

Feb 2026 – Current

Tempered AI

- Architected and built a three-layer LLM document extraction routing system on Azure Databricks, integrating cost-aware model selection, data quality preflight checks, and evidence-based hallucination detection to enable large-scale structured extraction from 5–200+ page insurance PDFs.
- Designed and implemented a self-escalating validation pipeline that reduced high-cost model invocations by 42% across 500 benchmark documents while improving extraction accuracy for 33 key fields. Developed a PreFlight document complexity classifier for three-tier model routing and introduced keyword coverage-based data routing to automatically detect and retry failed PDF extractions. Achieved a 30% cost reduction versus an all-o4-mini baseline with only a 3% increase in latency.
- Built a hybrid PDF processing pipeline combining Camelot/Tabula table extraction, layout-aware page filtering, and geometry-based pseudo-table detection. Designed a Carrier × LOB hierarchical YAML configuration framework and implemented CI/CD deployment using Azure Functions and GitHub Actions. Improved production Evidence Accuracy from 71% to 86% while maintaining processing costs at approximately \$0.14 per 1,000 pages.

Data Science & Data Analyst Intern— Health intelligence

Jan 2025 – Apr 2025

Region of Peel

- Develop data processing workflows, and the validation of forecasting models for public health facilities with Multi-variable Linear Regression and seasonal ARIMA.
- Provided actionable business insights based on model results to support data-driven decision-making in public health strategies.
- Completed model validation using MSE on test set and proposed workforce recruitment optimization recommendations based on forecasting results.

Machine Learning Intern

Sept 2024 – Dec 2024

Environment and Climate Change Canada

- Assist researchers in a scientific machine learning project, implement complex algorithms in Pytorch
- Implement and compare Transformer/Graph neural network Encoder-Decoder Model with Neural ODE Processor for Weather Forecasting train and test model using pytorch-Lightning
- Tested and debugged model training using memorization tests and unit tests to ensure robustness

Machine Learning Intern

May 2023 – Aug 2023

Institut Pasteur

- Developed and implemented K-means clustering algorithms to segment high- and low-fluorescence regions for detection of ridge-like spatial patterns in time-series image data.
- Designed and applied piecewise linear regression models to approximate ridge-like patterns and quantitatively measure velocity changes over time.

Education

University of Waterloo

Waterloo, Ontario

Master of Mathematics, Applied Mathematics

Sept. 2023 – Dec. 2025

University of Manitoba

Winnipeg, Manitoba

Bachelor of Science, Applied Math & Statistics

Sept. 2019 – May 2023

- Databricks Machine learning professional certificate, Databricks Data Engineering professional certificate
- Deep Learning, Machine Learning, NLP, Python, Pytorch, PySpark, Generative AI, R, Tableau, Java, Docker, ML model deployment, Matlab, MySQL, AWS Sagemaker, S3, Lambda, ECR
- Philosophia Mathematica Prize in Applied Mathematics (2022); Award for Highest Standing in Science Faculty (2023)
- X. Xu, Francis J, Poulin, Sue Ann Campbell, “Turing Instability of a Closed Nutrients-Phytoplankton-Zooplankton Model with Time Delay Effect,” *Mathematical Biosciences* (under review)

Project Experience

Clash Royale Game AI agent with Reinforcement Learning [Blog Link]

April 2025 - Aug 2025

- Apply visual fusion (YOLOv8, cnocr, resnet) for offline dataset feature extraction with data pipeline automation
- Train a decision transformer model for policy making through AWS sagemaker
- Deploy visual fusion and policy network on the android phone to play real-time game